

Section B

RaceDay - API Endpoint Plan

Base URL

<https://api.raceday.com/api>

Authentication

All protected endpoints require a JWT token in the Authorization header:

Authorization: Bearer {token}

API Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
-----	-----	-----	-----	-----	-----
POST	/auth/register	Register a new user	Public	{ "email", "password", "full_name", "role", "date_of_birth", "phone" }	201 Created: { "user_id", "message" }
POST	/auth/login	Login and receive JWT token	Public	{ "email", "password" }	200 OK: { "token", "user_id", "role" }
GET	/users/profile	Get current user profile	Authenticated	None	200 OK: User profile data
PUT	/users/profile	Update current user profile	Authenticated	{ "full_name", "date_of_birth", "phone", "emergency_contact" }	200 OK: Updated user profile
GET	/users/me	Get current user details	Authenticated	None	200 OK: { "user_id", "email", "full_name", "role" }
POST	/events	Create a new event	Organiser	{ "name", "description", "event_type", "start_date", "end_date", "location", "distance_km" }	201 Created: Event object
GET	/events	List all events (with filters)	Public	Query params: ?event_type, ?location, ?is_active	200 OK: Array of events
GET	/events/{eventId}	Get event by ID	Public	None	200 OK: Event object with categories
PUT	/events/{eventId}	Update an existing event	Organiser	{ "name", "description", "start_date", "end_date", "location", "is_active" }	200 OK: Updated event
DELETE	/events/{eventId}	Delete an event (soft delete)	Organiser	None	204 No Content

| POST | /categories | Add a category to an event | Organiser | { "event_id", "name", "min_age", "max_age", "entry_fee", "start_time" } | 201 Created: Category object |

| GET | /categories/{categoryId} | Get category by ID | Public | None | 200 OK: Category object |

| PUT | /categories/{categoryId} | Update a category | Organiser | { "name", "min_age", "max_age", "entry_fee", "start_time" } | 200 OK: Updated category |

| DELETE | /categories/{categoryId} | Delete a category | Organiser | None | 204 No Content |

| POST | /enrolments | Participant enrolls in an event | Participant | { "event_id", "category_id" } | 201 Created: Enrolment object with bib_number |

| GET | /enrolments | Get all enrolments for current participant | Participant | None | 200 OK: Array of enrolments with event details |

| GET | /enrolments/event/{eventId} | Get all enrolments for a specific event | Organiser | None | 200 OK: Array of enrolments |

| PUT | /enrolments/{enrolmentId} | Update enrolment (e.g., cancel) | Participant/Organiser | { "status" } | 200 OK: Updated enrolment |

| DELETE | /enrolments/{enrolmentId} | Cancel enrolment | Participant/Organiser | None | 204 No Content |

| POST | /results | Upload result for a participant | Organiser | { "enrolment_id", "finish_time", "position", "pace", "is_disqualified" } | 201 Created: Result object |

| GET | /results/event/{eventId} | Get all results for an event | Public | None | 200 OK: Array of results |

| GET | /results/participant | Get all results for current participant | Participant | None | 200 OK: Array of results |

| GET | /results/{resultId} | Get result by ID | Public | None | 200 OK: Result object |

| PUT | /results/{resultId} | Update a result | Organiser | { "finish_time", "position", "pace", "is_disqualified" } | 200 OK: Updated result |

| DELETE | /results/{resultId} | Delete a result | Organiser | None | 204 No Content |

Detailed Endpoint Specifications

1. Register User

Endpoint: POST /api/auth/register

Request Body:

{

```
"email": "participant@email.com",
"password": "SecurePassword123",
"full_name": "John Doe",
"role": "Participant",
"date_of_birth": "1990-05-15",
"phone": "+27123456789"
}
```

Response (201 Created):

```
{
  "user_id": 5,
  "message": "User registered successfully"
}
```

Response (400 Bad Request):

```
{
  "error": "Email already exists"
}
```

2. Login User

Endpoint: POST /api/auth/login

Request Body:

```
{
  "email": "participant@email.com",
  "password": "SecurePassword123"
}
```

Response (200 OK):

```
{
```

```
"token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
"user_id": 5,
"role": "Participant"
}
```

Response (401 Unauthorized):

```
{
  "error": "Invalid email or password"
}
```

3. Get Current User Profile

Endpoint: GET /api/users/profile

Headers:

Authorization: Bearer {token}

Response (200 OK):

```
{
  "user_id": 5,
  "email": "participant@email.com",
  "full_name": "John Doe",
  "role": "Participant",
  "date_of_birth": "1990-05-15",
  "phone": "+27123456789",
  "emergency_contact_name": "Jane Doe",
  "emergency_contact_phone": "+27123456790",
  "medical_conditions": "None"
}
```

4. Update User Profile

Endpoint: PUT /api/users/profile

Headers:

Authorization: Bearer {token}

Request Body:

```
{  
  "full_name": "John Smith",  
  "date_of_birth": "1990-05-15",  
  "phone": "+27123456788",  
  "emergency_contact_name": "Jane Smith",  
  "emergency_contact_phone": "+27123456789",  
  "medical_conditions": "Asthma"  
}
```

Response (200 OK):

```
{  
  "user_id": 5,  
  "email": "participant@email.com",  
  "full_name": "John Smith",  
  "role": "Participant",  
  "date_of_birth": "1990-05-15",  
  "phone": "+27123456788",  
  "emergency_contact_name": "Jane Smith",  
  "emergency_contact_phone": "+27123456789",  
  "medical_conditions": "Asthma"  
}
```

5. Get Current User Details

Endpoint: GET /api/users/me

Headers:

Authorization: Bearer {token}

Response (200 OK):

```
{  
  "user_id": 5,  
  "email": "participant@email.com",  
  "full_name": "John Smith",  
  "role": "Participant"  
}
```

6. Create Event

Endpoint: POST /api/events

Headers:

Authorization: Bearer {token}

Request Body:

```
{  
  "name": "Cape Town Marathon 2026",  
  "description": "The iconic Cape Town Marathon event",  
  "event_type": "Running",  
  "start_date": "2026-10-15T06:00:00",  
  "end_date": "2026-10-15T18:00:00",  
  "location": "Cape Town Stadium, Cape Town",  
  "distance_km": 42.2  
}
```

Response (201 Created):

```
{
  "event_id": 4,
  "organiser_id": 1,
  "name": "Cape Town Marathon 2026",
  "description": "The iconic Cape Town Marathon event",
  "event_type": "Running",
  "start_date": "2026-10-15T06:00:00",
  "end_date": "2026-10-15T18:00:00",
  "location": "Cape Town Stadium, Cape Town",
  "distance_km": 42.2,
  "is_active": true,
  "created_at": "2026-09-04T10:00:00"
}
```

7. List All Events

Endpoint: GET /api/events?event_type=Running&location=Cape%20Town&is_active=true

Query Parameters:

event_type - Filter by event type (Running, Walking, Cycling)

location - Filter by location

is_active - Filter by active status

Response (200 OK):

```
[
  {
    "event_id": 1,
    "name": "Cape Town Marathon 2026",
    "description": "The iconic Cape Town Marathon event",
    "event_type": "Running",
```

```
"start_date": "2026-10-15T06:00:00",
"end_date": "2026-10-15T18:00:00",
"location": "Cape Town Stadium, Cape Town",
"distance_km": 42.2,
"is_active": true
},
{
  "event_id": 2,
  "name": "Cape Town 10K",
  "description": "Shorter distance run",
  "event_type": "Running",
  "start_date": "2026-10-16T07:00:00",
  "end_date": "2026-10-16T12:00:00",
  "location": "Sea Point Promenade, Cape Town",
  "distance_km": 10.0,
  "is_active": true
}
]
```

8. Get Event by ID

Endpoint: GET /api/events/{eventId}

Response (200 OK):

```
{
  "event_id": 1,
  "organiser_id": 1,
  "name": "Cape Town Marathon 2026",
  "description": "The iconic Cape Town Marathon event",
  "event_type": "Running",
  "start_date": "2026-10-15T06:00:00",
```



```
"end_date": "2026-10-15T18:00:00",
"location": "Cape Town Stadium, Cape Town",
"distance_km": 42.2,
"is_active": true,
"created_at": "2026-09-01T10:00:00",
"categories": [
  {
    "category_id": 1,
    "name": "Elite Men",
    "min_age": 18,
    "max_age": 40,
    "entry_fee": 500.00,
    "start_time": "06:00:00"
  },
  {
    "category_id": 2,
    "name": "Elite Women",
    "min_age": 18,
    "max_age": 40,
    "entry_fee": 500.00,
    "start_time": "06:05:00"
  }
]
```

Response (404 Not Found):

```
{
  "error": "Event not found"
}
```

9. Update Event

Endpoint: PUT /api/events/{eventId}

Headers:

Authorization: Bearer {token}

Request Body:

```
{
  "name": "Cape Town Marathon 2026 - Updated",
  "description": "Updated description",
  "start_date": "2026-10-15T06:30:00",
  "end_date": "2026-10-15T19:00:00",
  "location": "Cape Town Stadium, Cape Town",
  "is_active": false
}
```

Response (200 OK):

```
{
  "event_id": 1,
  "organiser_id": 1,
  "name": "Cape Town Marathon 2026 - Updated",
  "description": "Updated description",
  "event_type": "Running",
  "start_date": "2026-10-15T06:30:00",
  "end_date": "2026-10-15T19:00:00",
  "location": "Cape Town Stadium, Cape Town",
  "distance_km": 42.2,
  "is_active": false,
  "created_at": "2026-09-01T10:00:00"
}
```

10. Delete Event

Endpoint: DELETE /api/events/{eventId}

Headers:

Authorization: Bearer {token}

Response: 204 No Content

11. Add Category to Event

Endpoint: POST /api/categories

Headers:

Authorization: Bearer {token}

Request Body:

```
{  
  "event_id": 1,  
  "name": "Junior Men",  
  "min_age": 16,  
  "max_age": 20,  
  "entry_fee": 250.00,  
  "start_time": "06:20:00"  
}
```

Response (201 Created):

```
{  
  "category_id": 11,  
  "event_id": 1,  
  "name": "Junior Men",
```

```
"min_age": 16,  
"max_age": 20,  
"entry_fee": 250.00,  
"start_time": "06:20:00",  
"created_at": "2026-09-04T10:00:00"  
}
```

12. Get Category by ID

Endpoint: GET /api/categories/{categoryId}

Response (200 OK):

```
{  
  "category_id": 1,  
  "event_id": 1,  
  "name": "Elite Men",  
  "min_age": 18,  
  "max_age": 40,  
  "entry_fee": 500.00,  
  "start_time": "06:00:00",  
  "created_at": "2026-09-01T10:00:00"  
}
```

13. Update Category

Endpoint: PUT /api/categories/{categoryId}

Headers:

Authorization: Bearer {token}

Request Body:

```
{  
  "name": "Elite Men (18-40)",  
  "min_age": 18,  
  "max_age": 40,  
  "entry_fee": 550.00,  
  "start_time": "06:00:00"  
}
```

Response (200 OK):

```
{  
  "category_id": 1,  
  "event_id": 1,  
  "name": "Elite Men (18-40)",  
  "min_age": 18,  
  "max_age": 40,  
  "entry_fee": 550.00,  
  "start_time": "06:00:00",  
  "created_at": "2026-09-01T10:00:00"  
}
```

14. Delete Category

Endpoint: DELETE /api/categories/{categoryId}

Headers:

Authorization: Bearer {token}

Response: 204 No Content

15. Enrol in Event

Endpoint: POST /api/enrolments

Headers:

Authorization: Bearer {token}

Request Body:

```
{  
  "event_id": 1,  
  "category_id": 2  
}
```

Response (201 Created):

```
{  
  "enrolment_id": 5,  
  "participant_id": 3,  
  "event_id": 1,  
  "category_id": 2,  
  "registration_date": "2026-09-04T10:00:00",  
  "status": "Registered",  
  "bib_number": 105  
}
```

16. Get Participant's Enrolments

Endpoint: GET /api/enrolments

Headers:

Authorization: Bearer {token}

Response (200 OK):

```
[
```

```
{
  "enrolment_id": 1,
  "event_id": 1,
  "event_name": "Cape Town Marathon 2026",
  "category_name": "Elite Women",
  "registration_date": "2026-09-01T10:00:00",
  "status": "Confirmed",
  "bib_number": 101,
  "finish_time": "03:45:30",
  "position": 15,
  "pace": "05:20"
},
{
  "enrolment_id": 2,
  "event_id": 2,
  "event_name": "Cape Town 10K",
  "category_name": "Open Women",
  "registration_date": "2026-09-02T10:00:00",
  "status": "Registered",
  "bib_number": 102
}
]
```

17. Get Enrolments for Event (Organiser Only)

Endpoint: GET /api/enrolments/event/{eventId}

Headers:

Authorization: Bearer {token}

Response (200 OK):

```
[
  {
    "enrolment_id": 1,
    "participant_id": 3,
    "participant_name": "Alice Johnson",
    "category_name": "Elite Women",
    "registration_date": "2026-09-01T10:00:00",
    "status": "Confirmed",
    "bib_number": 101
  },
  {
    "enrolment_id": 3,
    "participant_id": 4,
    "participant_name": "Bob Williams",
    "category_name": "Elite Men",
    "registration_date": "2026-09-02T10:00:00",
    "status": "Confirmed",
    "bib_number": 103
  }
]
```

18. Update Enrolment

Endpoint: PUT /api/enrolments/{enrolmentId}

Headers:

Authorization: Bearer {token}

Request Body:

```
{
  "status": "Cancelled"
}
```



```
}
```

Response (200 OK):

```
{  
  "enrolment_id": 2,  
  "participant_id": 3,  
  "event_id": 2,  
  "category_id": 5,  
  "registration_date": "2026-09-02T10:00:00",  
  "status": "Cancelled",  
  "bib_number": 102  
}
```

19. Cancel Enrolment

Endpoint: DELETE /api/enrolments/{enrolmentId}

Headers:

Authorization: Bearer {token}

Response: 204 No Content

20. Upload Result

Endpoint: POST /api/results

Headers:

Authorization: Bearer {token}

Request Body:

```
{
```

```
"enrolment_id": 1,  
"finish_time": "03:45:30",  
"position": 15,  
"pace": "05:20",  
"is_disqualified": false  
}
```

Response (201 Created):

```
{  
  "result_id": 3,  
  "enrolment_id": 1,  
  "finish_time": "03:45:30",  
  "position": 15,  
  "pace": "05:20",  
  "is_disqualified": false,  
  "created_at": "2026-09-04T10:00:00"  
}
```

21. Get Results for Event

Endpoint: GET /api/results/event/{eventId}

Response (200 OK):

```
[  
  {  
    "result_id": 1,  
    "enrolment_id": 1,  
    "participant_name": "Alice Johnson",  
    "category_name": "Elite Women",  
    "finish_time": "03:45:30",  
    "position": 15,
```

```
    "pace": "05:20",
    "is_disqualified": false
  },
  {
    "result_id": 2,
    "enrolment_id": 3,
    "participant_name": "Bob Williams",
    "category_name": "Elite Men",
    "finish_time": "02:45:15",
    "position": 5,
    "pace": "04:15",
    "is_disqualified": false
  }
]
```

22. Get Participant's Results

Endpoint: GET /api/results/participant

Headers:

Authorization: Bearer {token}

Response (200 OK):

```
[
  {
    "result_id": 1,
    "event_id": 1,
    "event_name": "Cape Town Marathon 2026",
    "category_name": "Elite Women",
    "finish_time": "03:45:30",
    "position": 15,
```

```
    "pace": "05:20",
    "is_disqualified": false
  }
]
```

23. Get Result by ID

Endpoint: GET /api/results/{resultId}

Response (200 OK):

```
{
  "result_id": 1,
  "enrolment_id": 1,
  "finish_time": "03:45:30",
  "position": 15,
  "pace": "05:20",
  "is_disqualified": false,
  "created_at": "2026-09-03T10:00:00"
}
```

24. Update Result

Endpoint: PUT /api/results/{resultId}

Headers:

Authorization: Bearer {token}

Request Body:

```
{
  "finish_time": "03:40:15",
  "position": 12,
```

```
"pace": "05:10",  
"is_disqualified": false  
}
```

Response (200 OK):

```
{  
  "result_id": 1,  
  "enrolment_id": 1,  
  "finish_time": "03:40:15",  
  "position": 12,  
  "pace": "05:10",  
  "is_disqualified": false,  
  "created_at": "2026-09-03T10:00:00"  
}
```

25. Delete Result

Endpoint: DELETE /api/results/{resultId}

Headers:

Authorization: Bearer {token}

Response: 204 No Content

Role-Based Access Control

Role: Organiser

Permissions: Create, update, delete events; Add, update, delete categories; View all enrolments for their events; Upload, update, delete results

Role: Participant

Permissions: Register for events; View own enrolments; Cancel own enrolments; View own results; Update own profile

Role: Public

Permissions: View events; View event details; View results; Register (create account)

HTTP Status Codes

200 OK - Request successful

201 Created - Resource created successfully

204 No Content - Request successful, no content returned

400 Bad Request - Invalid request parameters

401 Unauthorized - Missing or invalid authentication token

403 Forbidden - Insufficient permissions

404 Not Found - Resource not found

409 Conflict - Resource already exists (e.g., duplicate email)

500 Internal Server Error - Server error

Error Response Format

All error responses follow this format:

```
{  
  "error": "Error message description",  
  "status": 400,  
  "timestamp": "2026-09-04T10:00:00"  
}
```

Endpoint Summary

Category: Authentication

Endpoints: 2

Description: Register and Login

Category: User Profile

Endpoints: 3

Description: Get, update, and view user details

Category: Event Management

Endpoints: 5

Description: Create, read, update, delete, list events

Category: Category Management

Endpoints: 4

Description: Create, read, update, delete categories

Category: Enrolment Management

Endpoints: 5

Description: Enrol, view, update, cancel enrolments

Category: Result Management

Endpoints: 6

Description: Upload, view, update, delete results

Total: 25 Endpoint